

Setting Up Zephyr Tracing (CTF + RAM Backend)

This guide assumes OpenOCD + GDB are already working on your ESP32-S3. If not, follow [docs/installing_ocd.md](#) first.

Prerequisites

- `trace_demo` built and running on the board
- USB-JTAG working (OpenOCD + GDB connect successfully)
- Trace Compass installed: Install from: <https://eclipse.dev/tracecompass/>

1) Start OpenOCD

Use the same command from [docs/installing_ocd.md](#):

```
openocd -f board/esp32s3-builtin.cfg -c "tcl port 6667"
```

2) Start GDB and attach

Run from the course root so the relative ELF path resolves and the trace dump lands in the same folder:

`C:\zephyr-esp32s3-workspace\zephyr-esp32s3-course`

```
c:/users/johno/zephyr-sdk-0.17.4/xtensa-espressif_esp32s3_zephyr-elf/bin/xtensa-espressif_esp32s3_zephyr-elf-gdb.exe .\build\trace_demo\zephyr\zephyr.elf
```

Then in GDB:

```
target extended-remote localhost:3333
```

3) Dump the RAM tracing buffer

Zephyr's RAM backend stores trace data in the `ram_tracing` array. Dump the full buffer size (default: 65536 bytes).

Create a **single data/** folder in the same directory where GDB is launched:

```
mkdir data
```

In GDB:

```
p/x &ram_tracing
set $ram_start = (char *)&ram_tracing
set $ram_end = $ram_start + 65536
dump binary memory data/channel0_0 $ram_start $ram_end
```

4) Add CTF metadata (same **data/** folder)

Trace Compass needs two files to decode our raw trace channel:

- **data/channel0_0**: the binary event stream dumped from target RAM.
- **data/metadata**: the CTF schema (event definitions, field layouts, types).

That Copy-Item command copies Zephyr's CTF metadata template into our trace folder so Trace Compass knows how to interpret bytes in **channel0_0**. Without metadata, the tool sees raw bytes but can't decode events like thread switches, semaphores, or our named markers.

Windows (PowerShell, run from the course root): **C:\zephyr-esp32s3-workspace\zephyr-esp32s3-course**

```
# If ZEPHYR_BASE is set:
Copy-Item -Force "$env:ZEPHYR_BASE\sys\target\target\tracing\ctf\tsdl\metadata"
"data\metadata"

# If ZEPHYR_BASE is not set, use the workspace Zephyr tree (one level up):
Copy-Item -Force "..\zephyr\sys\target\target\tracing\ctf\tsdl\metadata"
"data\metadata"
```

Linux/macOS:

```
mkdir -p data
cp "$ZEPHYR_BASE/sys\target\target\tracing\ctf\tsdl\metadata" data/metadata
```

Ensure **data/** contains:

- **metadata**
- **channel0_0**

5) Open in Trace Compass

1. File → Open Trace
2. Select the **data/** folder
3. View scheduling, ISR, and blocking patterns